COS30049 – Computing Technology Innovation Project

Week 3 - Monitoring, Controlling, and Closure in Project Management & Usability of a Web Project

(Lecture – 01)

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Acknowledgement of Country

We respectfully acknowledge the Wurundjeri People of the Kulin Nation, who are the Traditional Owners of the land on which Swinburne's Australian campuses are located in Melbourne's east and outer-east, and pay our respect to their Elders past, present and emerging.

We are honoured to recognise our connection to Wurundjeri Country, history, culture, and spirituality through these locations, and strive to ensure that we operate in a manner that respects and honours the Elders and Ancestors of these lands.

We also respectfully acknowledge Swinburne's Aboriginal and Torres Strait Islander staff, students, alumni, partners and visitors.

We also acknowledge and respect the Traditional Owners of lands across Australia, their Elders, Ancestors, cultures, and heritage, and recognise the continuing sovereignties of all Aboriginal and Torres Strait Islander Nations.



Objectives of Today

Project Monitoring and Controlling

- KPIs
- Change Control
- Quality Control
- Risk Control

Project Closure

- Acceptance Criteria

Usability of a Web Project

- Usability Principles
- Usability Testing
- Real Word Examples
- Additonal Tips for Design

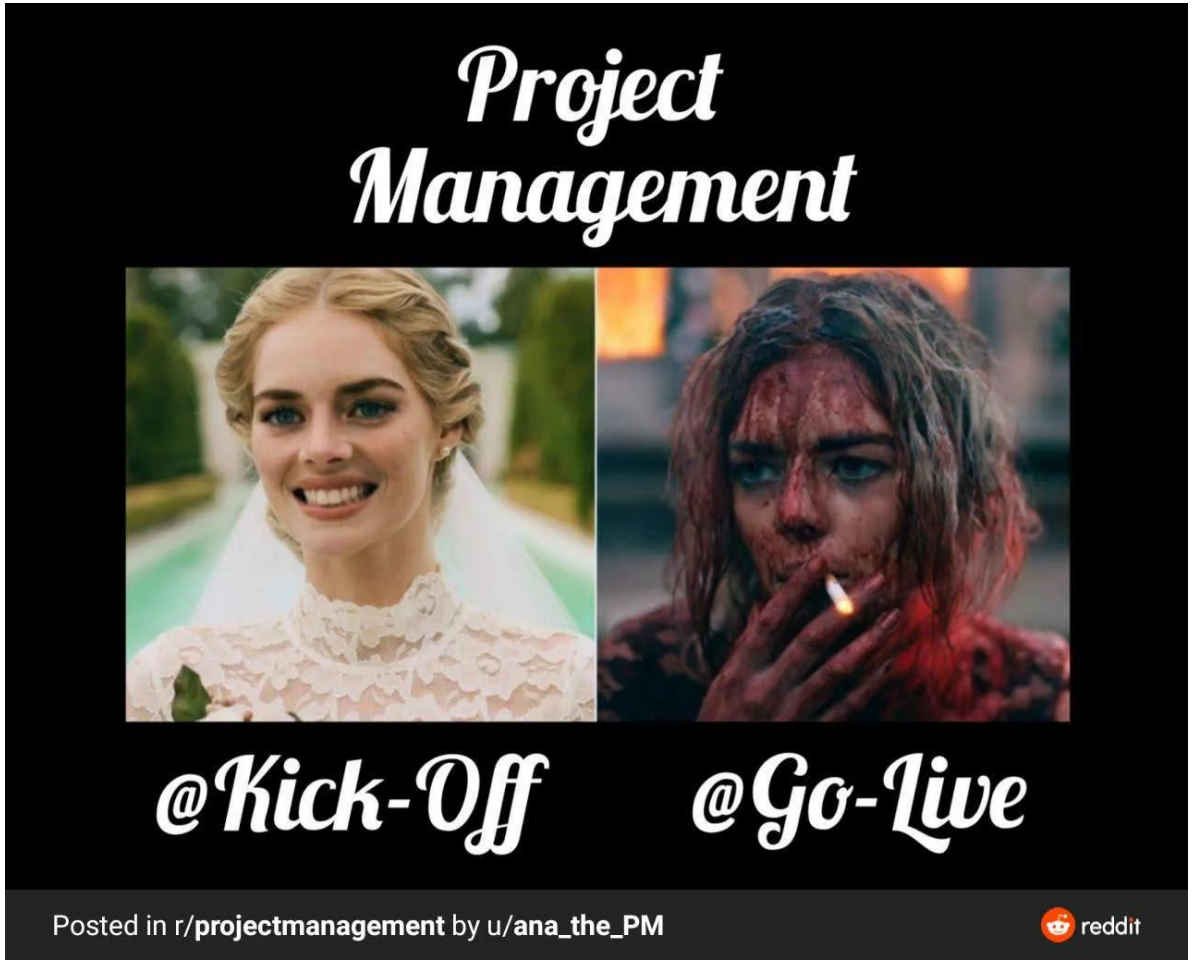
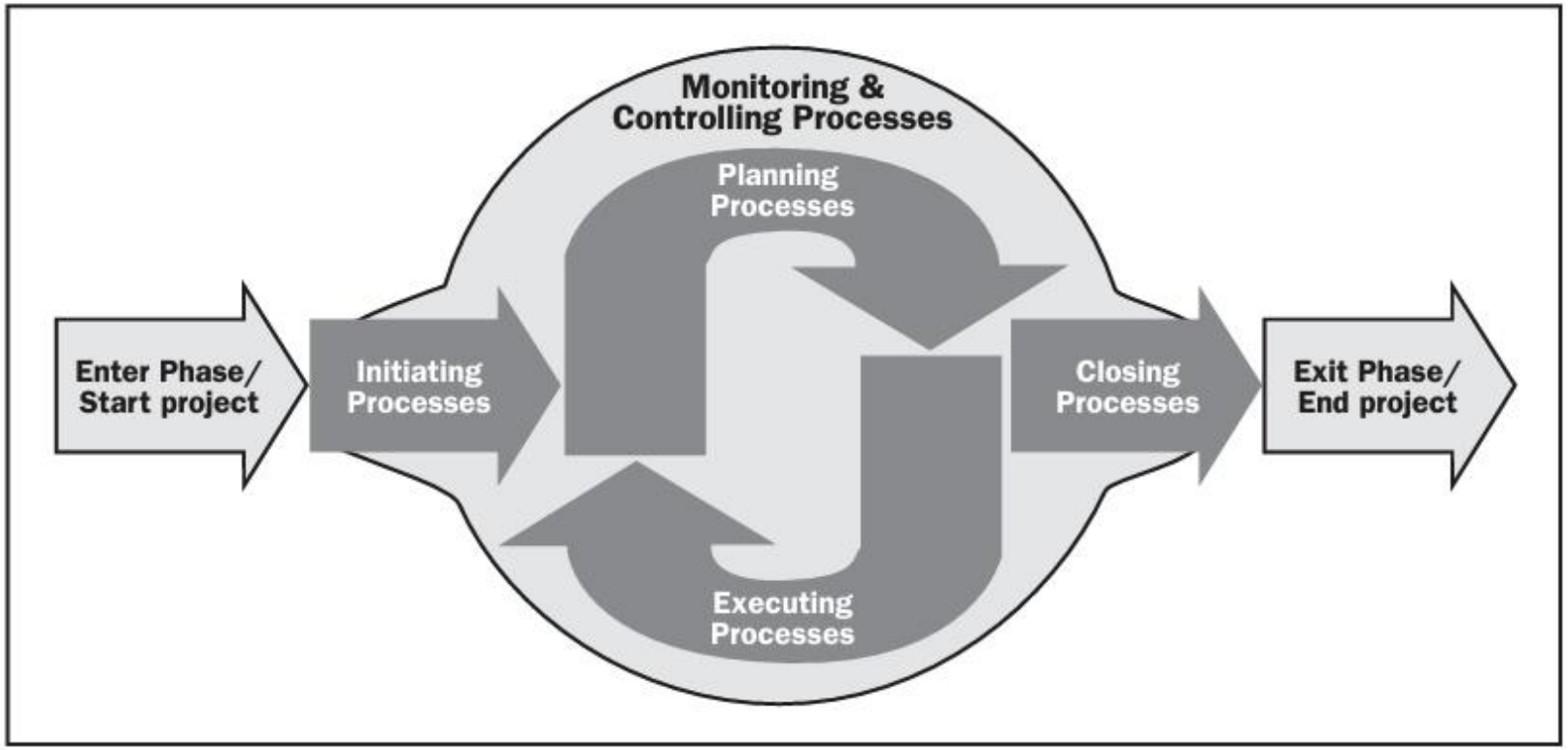


Image from [Reddit]

Project lifecycle

1. Initiation
2. Planning
3. Execution
4. **Monitoring**
5. closure



Project Management Process,

Project Management Institute, 2001, p.40

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Project Monitoring and Controlling



Photo by [Jayasahan Hansana] on [Unsplash]

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Case Study

The Denver International Airport Baggage System

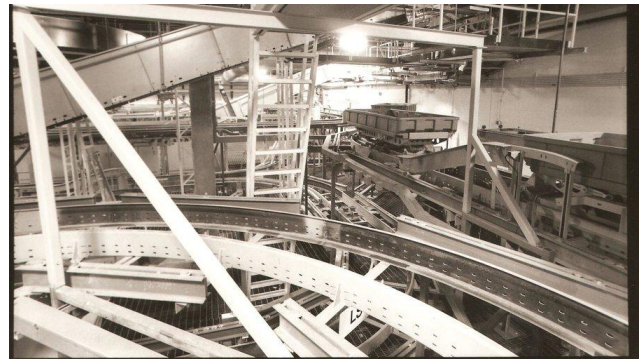


Image Source: [Biz Journals]

- **Overview**: The automated baggage handling system at Denver International Airport (DIA) was intended to be a state-of-the-art system to improve efficiency and handling time for baggage.
- **Project Start**: The project began in the early 1990s with high expectations and significant investment.
- **Goal**: The system was designed to automate baggage handling, reduce transfer times, and improve accuracy in baggage delivery.
- **Investment**: The project had a budget of approximately $193 million.
- **Complexity**: The system included 300 computers, 3,100 standard conveyors, 5,000 electric motors, and 56 laser arrays to route baggage.

More info:
Denver International Airport Baggage Handling System – An illustration of ineffectual decision making

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Case Study

The Denver International Airport Baggage System



Image Source: [Biz Journals]

Key Factors for Failure:
• **Inadequate Monitoring**: There was insufficient tracking of the project's progress, leading to missed deadlines and milestones.
• **Poor Risk Management**: Risks were not properly identified or managed, resulting in unforeseen issues.
• **Communication Breakdowns**: Lack of effective communication between stakeholders and project teams led to misunderstandings and delays.
• **Scope Creep**: Changes to the project scope were not adequately controlled, leading to increased complexity and delays.
• **Cost Overruns**: The project experienced significant cost overruns due to poor budget management and unforeseen expenses.

Outcome:
• **Abandonment**: The automated baggage handling system was eventually abandoned after significant delays and cost overruns.
• **Financial Loss**: The failure resulted in a considerable financial loss, with final costs estimated at over $560 million.
• **Reputation Damage**: The project failure damaged the reputations of those involved and became a case study in project management failure.

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Importance of Monitoring and Controlling

The monitoring and controlling phase is a crucial aspect of project management, responsible for tracking, reviewing, and regulating the progress and performance of a project. This phase ensures that the project stays on track, adheres to the plan, and achieves its objectives

- Ensures project stays on track
  - Regular progress review
  - Comparison against project baselines
- Helps manage risks effectively
  - Early identification of issues
  - Proactive mitigation strategies
- Enables informed decision-making
  - Data-driven insights
  - Timely adjustments
- Facilitates stakeholder communication and satisfaction
  - Regular updates
  - Transparent processes

Impact on Project Success:

- Effective management ensures the project meets its objectives and is delivered on time and within budget.

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Introduction to Monitoring and Controlling

Overview:

Monitoring and controlling in project management involve the systematic tracking, reviewing, and regulating of project progress and performance to ensure that project objectives are achieved

Purpose:

- Track, review, and regulate project progress and performance.
- Identify areas where changes to the plan are required.
- Ensure project objectives are met.

Key Concepts for This Phase in This Lecture:

1. Key Performance Indicators
2. Change Control
3. Quality Control
4. Risk management

Why Do We Need to Mornitor and Control phase in Project Management?

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Monitor Project Performance and Progress

Key Metrics Overview:

- Focus on critical Key Performance Indicators (KPIs) that provide insights into project performance.



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01 - Key Performance Indicators (KPIs)

Schedule Variance (SV):

- Definition: Measures the difference between planned and actual progress.
- Formula: SV = Earned Value (EV) - Planned Value (PV)
- Purpose: Identifies whether the project is ahead or behind schedule.

Cost Variance (CV):

- Definition: Measures the difference between budgeted cost and actual cost
- Formula: CV = Earned Value (EV) - Actual Cost (AC)
- Purpose: Indicates whether the project is under or over budget.

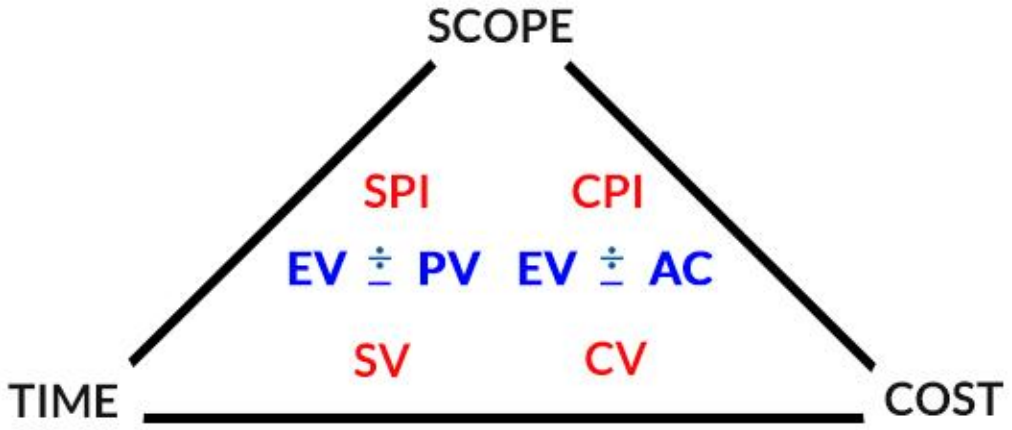


Image Source: ["Earned Value Management (EVM) for Agile Software Projects," CodeTiburon]

Schedule Performance Index (SPI):

- Definition: Ratio of earned value to planned value.
- Formula: SPI = EV / PV
- Purpose: Provides insight into time utilization efficiency.

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01 - Key Performance Indicators (KPIs)

Cost Performance Index (CPI):

• Definition: Ratio of earned value to actual cost.

• Formula: CPI = EV / AC

• Purpose: Measures cost efficiency of budgeted resources.

• Quality Metrics:

• Definition: Measures the quality of project deliverables.

• Purpose: Ensures outputs meet required standards.

• Scope Metrics:

• Definition: Tracks the scope of work completed versus planned.

• Purpose: Ensures that the project stays within its defined scope.

• Risk Metrics:

• Definition: Measures the identification, analysis, and mitigation of project risks.

• Purpose: Ensures that potential risks are managed and minimized.

• Stakeholder Metrics:

• Definition: Tracks stakeholder engagement and satisfaction.

• Purpose: Ensures that stakeholder needs and expectations are met.

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01 - Key Performance Indicators (KPIs)

Tools:	
Gantt charts	Visual representation of the project schedule
Dashboards	Real-time project metrics
Status reports	Periodic updates on project progress

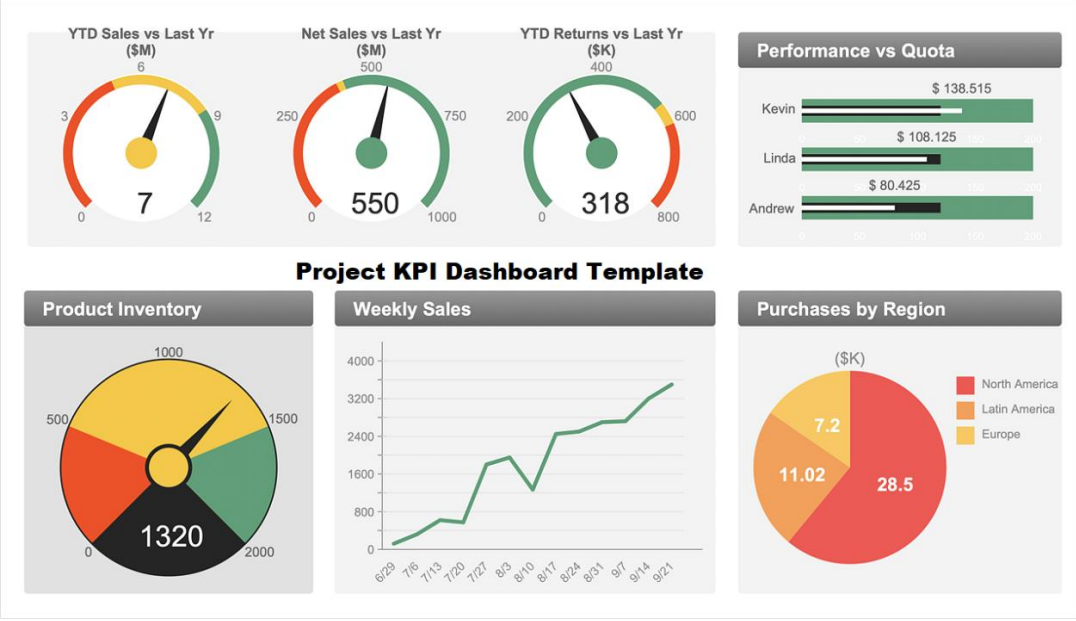


Image from: [Muhammad Omair, "Project Management KPI Dashboard," LinkedIn]

Status reports

Dashboard

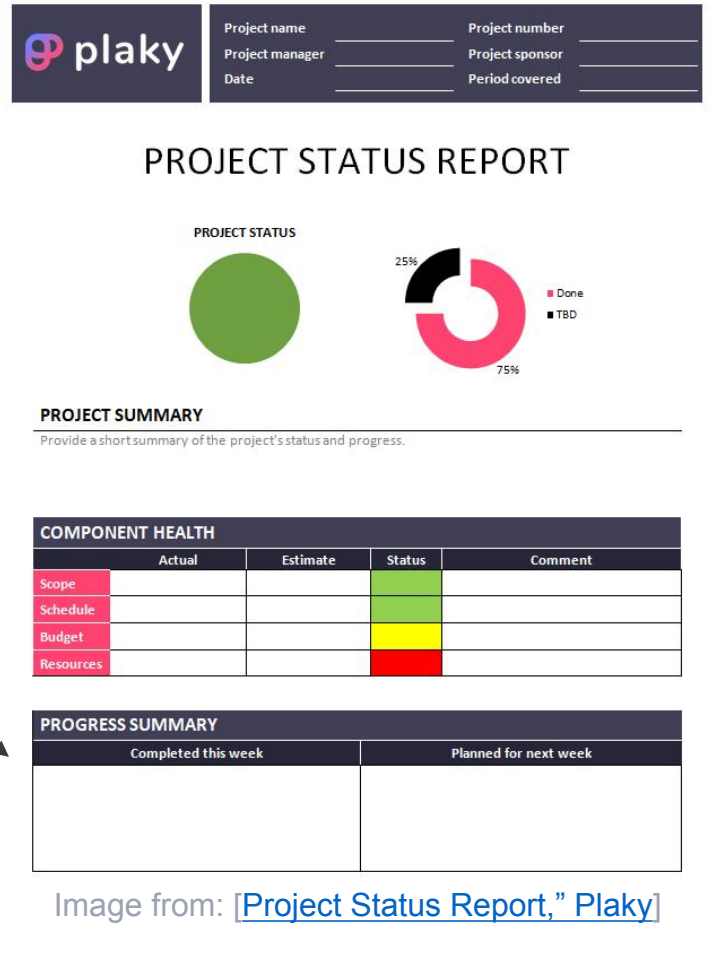


Image from: [Project Status Report," Plaky]

02 - Change Control (A1)

We need a plan not only managing the project, but also for managing the scope of the project.

- Change Control Board (CCB):

A group of stakeholders responsible for reviewing, evaluating, approving, or rejecting changes to a project.

- Change Log:

The Change Log serves as a formal record of all change requests, tracking their status from submission through resolution.

The CCB ensures that changes are evaluated and approved in a structured manner, while the Change Log maintains a transparent record of all changes, supporting informed decision-making and project success



Image Source: by [Project Management Memes," Plaky]

Change Control - Change Control Board (CCB)



Image Source: [Change Control Board (wallstreetmojo.com)]

- **Definition**: A Change Control Board (CCB) is a group of stakeholders responsible for reviewing, evaluating, approving, or rejecting changes to a project.
- **Purpose**: The CCB ensures that all changes are systematically considered and aligned with the project's objectives, scope, budget, and timeline.

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Change Control - Change Control Board (CCB) (A1)

Roles included in CCB:

- Project Manager, Project Sponsor, Technical Lead/Architect, Quality Assurance (QA) Manager, Financial Manager, Business Analyst, Stakeholder Representatives

Responsibilities:

- Review Process: Evaluates change requests based on their impact on project scope, schedule, budget, and quality.
- Decision Making: Approves, rejects, or defers change requests, ensuring that only beneficial changes are implemented.

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Change Control - Change Log (A1)

- **Definition**: A Change Log is a document or tool used to record all change requests, their status, and their impact on the project.
- **Purpose**: Provides a historical record of all changes, facilitating tracking, accountability, and communication**.**

Key Components of a Change Log:

- Change ID: Unique identifier for each change request.
- Description: Brief summary of the change request.
- Requestor: Name of the person or entity requesting the change.
- Date Submitted: When the change request was submitted.
- Status: Current status of the change (e.g., pending, approved, rejected, implemented).
- Impact Analysis: Assessment of how the change affects project scope, schedule, budget, and quality.
- Decision: Outcome of the change request (approved/rejected) by the CCB.
- Comments: Additional notes or comments related to the change request.

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Change Control - Change Log

Change Request ID	Date of Request	Change Description	Requester	Impact Summary	Status	Decision Date	Decision Made By	Notes
CR-001	2024-08-10	Extend project timeline by two weeks	Project Manager	Minor impact on budget, no impact on scope	Approved	2024-08-12	CCB	Update project schedule; notify stakeholders
CR-002	2024-08-11	Add a new feature to the software module	Client	Significant increase in scope and budget	Rejected	2024-08-13	CCB	Reevaluate in next project phase, not feasible now
CR-003	2024-08-13	Change supplier for key materials due to cost	Procurement Team	Slight delay expected, potential cost savings	Approved	2024-08-15	CCB	Update procurement plan, inform supplier of changes
CR-004	2024-08-14	Adjust resource allocation for the testing phase	Testing Lead	No impact on budget, slight schedule adjustment	Approved	2024-08-16	CCB	Redistribute resources and update project plan
CR-005	2024-08-15	Reduce the scope of the documentation requirements	Documentation	Minor reduction in scope, slight cost reduction	Approved	2024-08-17	CCB	Update documentation guidelines, communicate to team

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Change Control - Change Log VS CCB

	Change Control Board (CCB)	Change Log
Role	Group of stakeholders who review, evaluate, approve, or reject change requests	Document or tool used to record and track all change requests and their statuses
Responsibilities	Evaluates the impact of changes on project scope, schedule, budget, and quality. • Makes decisions to approve, reject, or defer changes. • Ensures alignment with project objectives and risk management.	Records all submitted change requests with unique IDs and descriptions. • Tracks the status of each change (pending, approved, rejected, implemented). • Provides a historical record of all changes and their impacts on the project.
Relationship	• CCB Uses Change Log: The CCB reviews and updates the Change Log to document decisions and track the status of change requests. • Change Log Supports CCB: The Change Log provides the CCB with the necessary information to make informed decisions and maintain accountability.	

Change Control - Change Log VS CCB (A1)

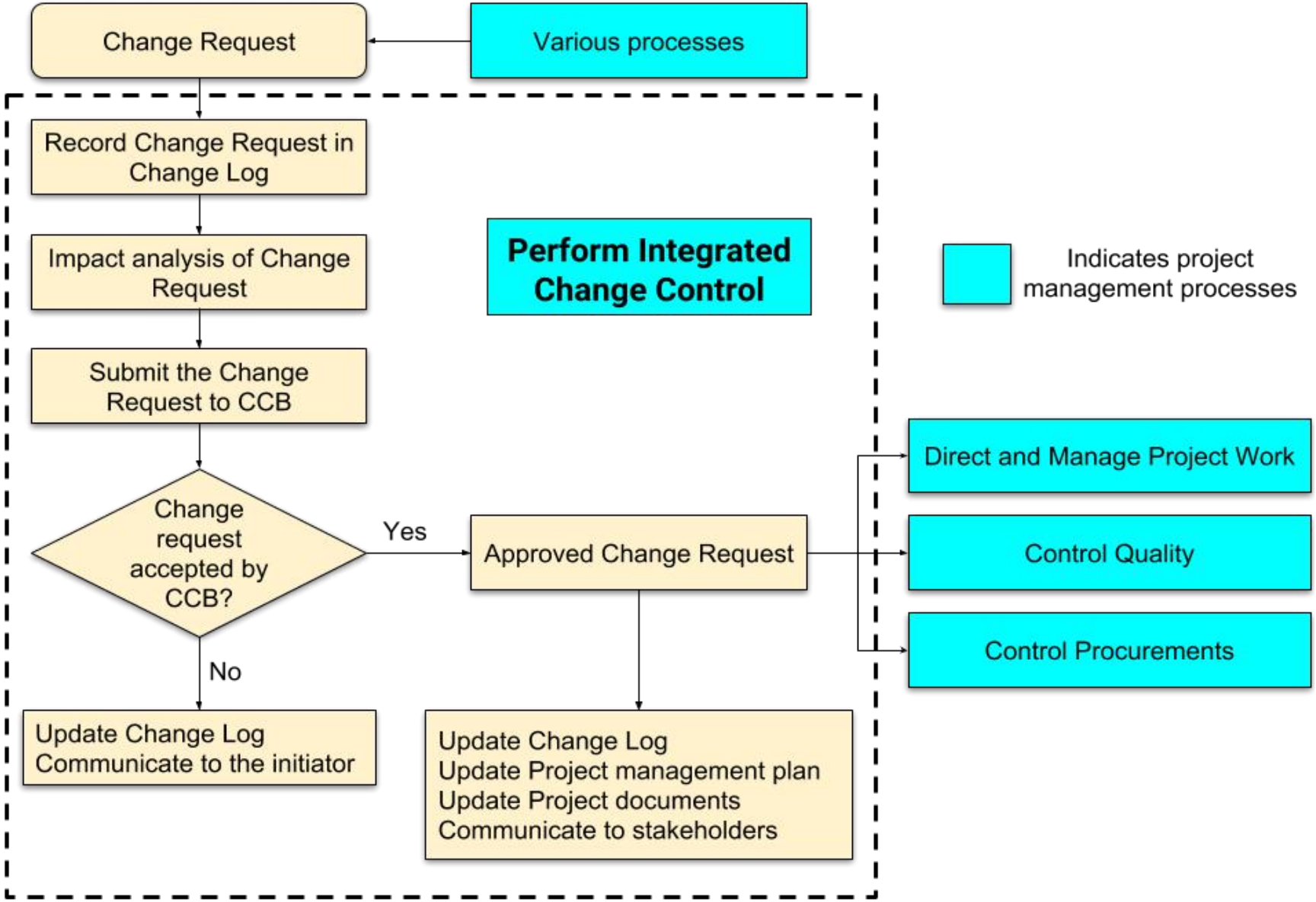


Image Source: [Perform Integrated Change Control by Manickavel Arumugam (pmdrill.com)]

03. Quality Control

Quality control involves monitoring and documenting the outcomes of quality-related activities to assess performance and recommend improvements where necessary.

Quality Assurance vs. Quality Control:

- Quality Assurance (QA): Focuses on ensuring that processes are followed consistently to maintain quality throughout the project. It is proactive and process-oriented.
- Quality Control (QC): Involves inspecting and testing project deliverables to ensure they meet the required quality standards. It is reactive and product-oriented.

In a project, poor quality can lead to... increased costs, delays, and customer dissatisfaction. How do you mitigate these risks?

What tools and techniques do you use to ensure consistent quality in your projects?

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Quality Control

Tools:	
Control Charts	**Purpose:** Monitor process stability and identify any variations that may affect the quality of the product or service. **Example:** A control chart might be used in manufacturing to track the consistency of product dimensions over time, ensuring that they stay within specified limits.
Checklists	**Purpose:** Ensure all required steps in a process are followed to maintain quality standards. **Example:** A checklist could be used in project management to confirm that all necessary approvals have been obtained before moving to the next project phase.
Pareto Diagrams	**Purpose:** Identify the most significant sources of defects or issues, often based on the Pareto principle (80/20 rule). **Example:** A Pareto diagram might show that 80% of product defects come from 20% of the possible causes, allowing teams to focus on the most critical areas for improvement.

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04. Risk Management (A1)

Project Risk:

A detailed description of potential uncertainties in the project scope, including deliverables, assumptions, and constraints.

➢ Types of Risks:

- External Risks: Market changes, legal issues, natural disasters.
- Internal Risks: Resource availability, technical failures, budget constraints.
- Technical Risks: Risks related to the technology being used.
- Operational Risks: Risks related to internal processes and procedures.

Risk Registers:

- To evaluate the effectiveness of the risk management processes in place

Risk Audits:

- Evaluating the effectiveness of risk management processes.



Image Source: by [Project Management Memes," Plaky]

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04. Risk Management – Risk Registration Plan (A1)

Writing a Risk Registration Plan:

A. Introduction:

- Define the purpose of the risk registration plan.
- Explain its importance in the overall project management process.

B. Steps to Create a Risk Registration Plan:

1. Risk Identification:

- List all potential risks identified during the planning phase.
- Use brainstorming sessions, expert interviews, and historical data to identify risks.

2. Risk Assessment:

- Evaluate the impact and likelihood of each risk.
- Prioritize risks based on their potential impact on the project.

3. Risk Response Planning:

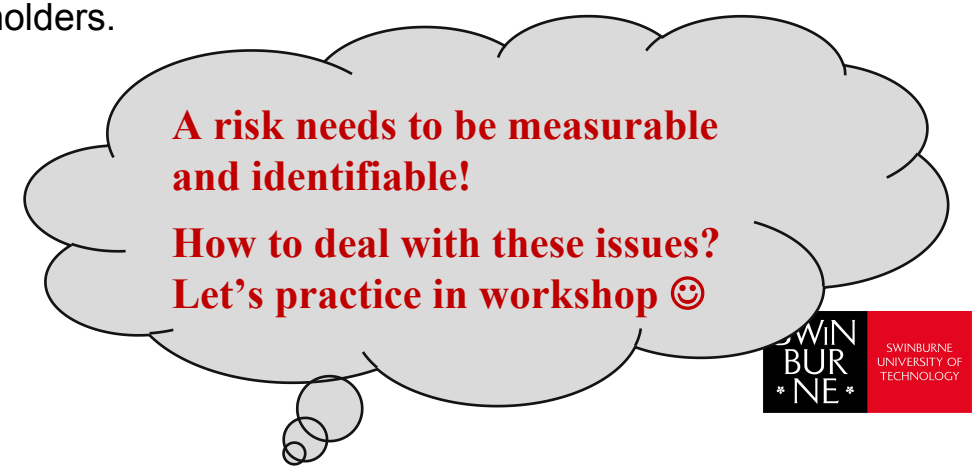
- Develop strategies for mitigating, accepting, transferring, or avoiding risks.
- Assign responsibility for each risk to a team member.

4. Risk Monitoring and Control:

- Establish procedures for monitoring risks throughout the project.
- Schedule regular reviews to reassess and update the risk register.

5. Documentation:

- Record all identified risks, assessments, and responses in the risk register.
- Ensure the risk register is accessible to all relevant stakeholders.



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04. Risk Management – Risk Register Diagram (A1)

Sr No	Date Raised	Risk Description	Likelihood of the risk	Impact of the risk occurs	Severity	Owner	Mitigating Action	Contingency Plan	Progress on Action	Status	Resource
1	20 Dec 2021	Project Sprints stories are not well defined	High	High	High	Product Owner	Complete the Project Charter having all the business cases and ensure requirements are gathered properly	Escalate to project Board	Project Charter should be rewritten with full business case and requirements	Open	Project Charter Template
2	22 Dec 2021	Project design and deliverables/milestones are incomplete.	High	High	High	Product Owner	Define the scope and deliverables in detail OR take help from experts	Assumptions and constraints are defined in the document. Remove high-risk items.	Workshops for training on how to create deliverables and milestones scheduled	In Progress	Assumptions Log
3	23 Dec 2021	Non- Competent Resources	Medium	Medium	Medium	Project Manager	Schedule training or coaching from experts	Escalate to the project sponsor	The project sponsor agrees to hold a briefing and make arrangements for new resources	In Progress	Resource Planning Sheet
4	24 Dec 2021	Delays in appointing Expert Consultant or Freelancer	Low	Low	Low	Project Manager	Communicate schedule early and check-in with consultants and contractors regularly.	Escalate to the project sponsor	Lead time to be built on the project schedule for each delay in onboarding	Open	Lead time
5	26 Dec 2021	Unplanned work	High	High	High	Project Manager	Check all the plans and scope of the project. Document all assumptions and communicate clearly to the project manager.	Escalate to the project manager and develop a plan to mitigate cost and schedule	Project manager to attend a meeting with the project sponsor	Open	Project Schedule

Note:
The diagram you saw here is a Risk Register diagram, which is a document that lists all identified risks for a project, along with details such as the likelihood and impact of each risk.

This is **different from a Risk Registration Plan**, which outlines the process and methodology for identifying, assessing, and managing risks.

Image Source:[Fahad Usmani, "Risk Register in Project Management," PM Study Circle]

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Project Closure

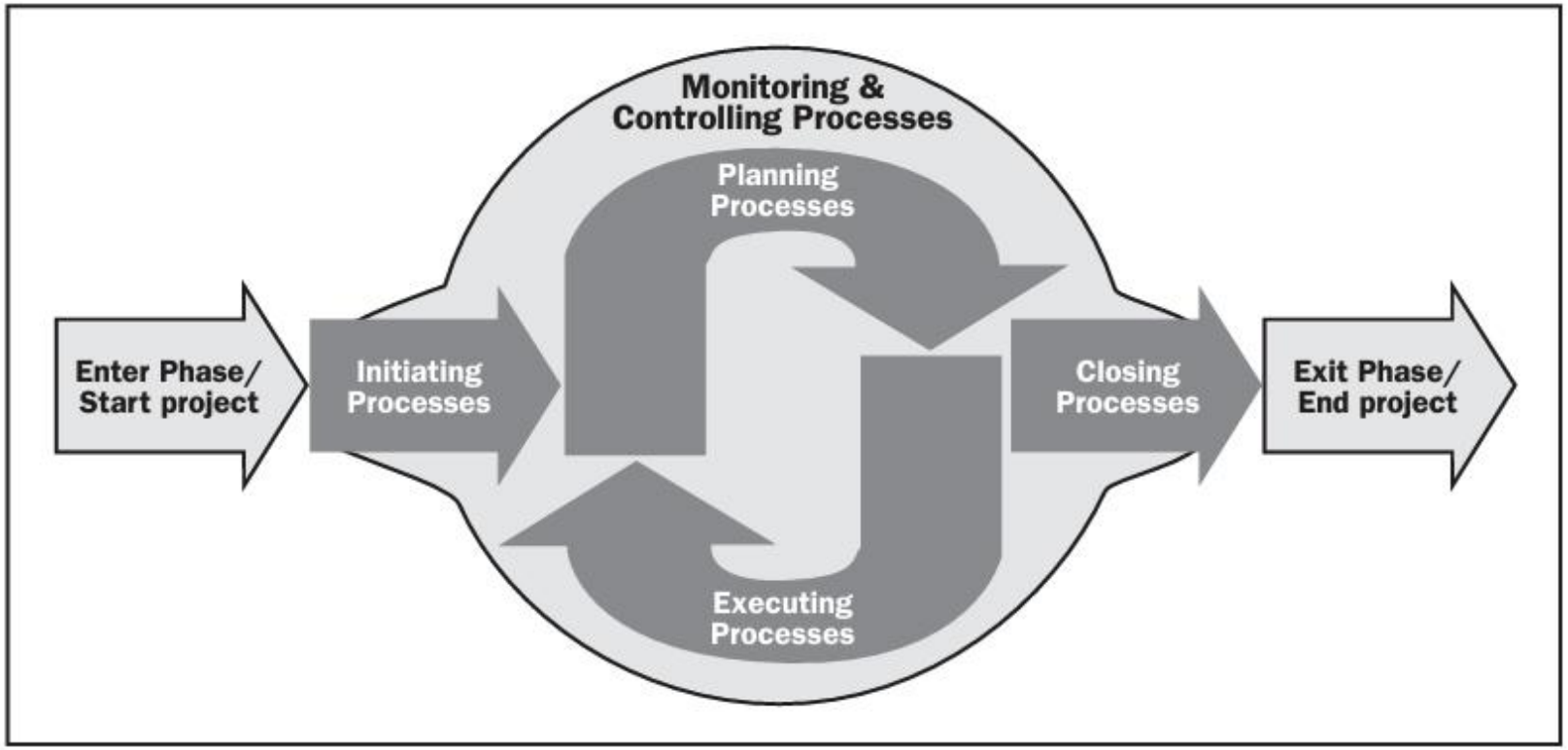


Image Source: by [imageflip]

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Project lifecycle

1. Initiation
2. Planning
3. Execution
4. Monitoring
5. **Closure**



Project Management Process,

Project Management Institute, 2001, p.40

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Introduction to Project Closure

What is Project Closure?

The process of finalizing all project activities, handing over deliverables, releasing project resources, and confirming project completion.

Key Steps:

- **Project Deliverables**: Ensure that all project outputs are completed and meet the required standards.
- **Acceptance Criteria (A1)**: Verify that the deliverables meet the predefined standards or requirements agreed upon with the client or stakeholders.
- **Obtain Formal Acceptance**: Secure formal approval from the client or stakeholders based on the acceptance criteria.
- **Document Final Deliverables**: Compile and archive all project documentation and final reports.
- **Release Project Resources**: Reassign or release the project team, budget, and any other resources used during the project.
- **Conduct Post-Project Review**: Evaluate the project's performance and identify lessons learned for future projects.

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Project Closure – Acceptance Criteria

What is Acceptance Criteria?

Predefined standards or requirements that deliverables must meet for them to be accepted by the client or stakeholders.

Purpose:

- Ensures project deliverables meet the expectations.
- Provides a clear basis for approval.

Examples:

- **Functionality**: “All functionalities must be tested and work as specified in the requirements document.”
- **Performance**: “The website must load in under 3 seconds on all major browsers.”
- **Usability**: “The application must pass usability testing with a minimum satisfaction score of 85% from user feedback.”

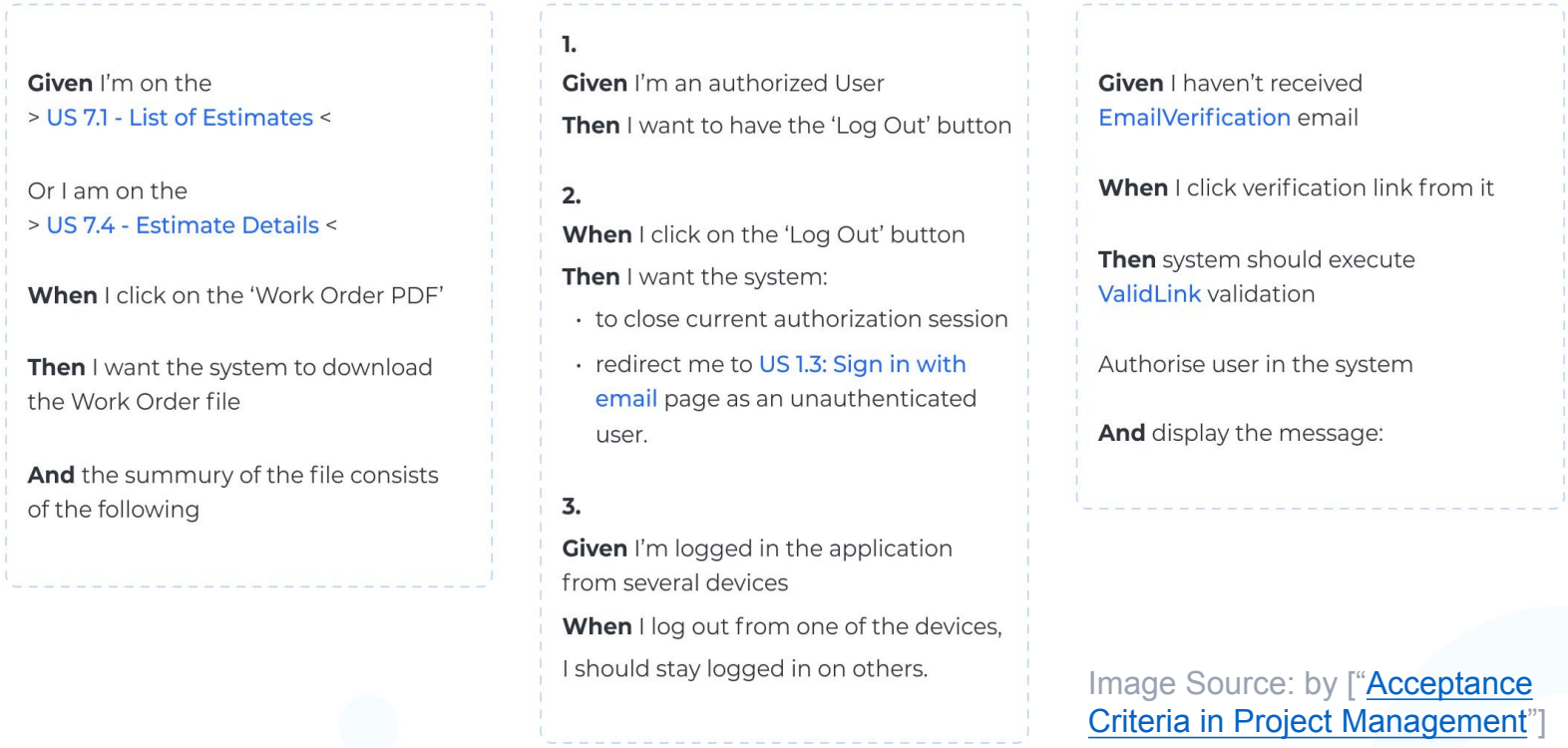


Image Source: by [“Acceptance Criteria in Project Management”]

It needs to be more detailed in practice; you can refer to the example in the picture above

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Project Closure – Acceptance Criteria (A1)

Some tips when writing AC:

- Always write a scenario name for each acceptance criteria, it won't take too much time but will bring your team a lot of benefits.
- Use simple English to be sure all the specs users will fully understand you
- A popular approach to writing acceptance criteria is the scenario-oriented **Given / When / Then approach**
  - **Given** is a clause that describes the state before the user entered the script.
  - **When** – something happens: the user takes some action.
  - **Then** – a description of how the system responds to user actions.

For example, if you create a website with two types of users (registered users and guests), the acceptance criteria will be written for the user story that defines requirements for the logged out user:

Example of acceptance criteria

Sign Up:
1. System user signs in with valid credentials

"**Given** I'm a logged-out system user
and I'm on the Sign-In page
When I fill in the "Username" and "Password" fields with my authentication credentials
and I click the Sign-In button
Then the system signs me in"

Given / When / Then helps reduce the time it takes to write test cases because when writing criteria, we describe the system's behaviour in advance.

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Project Closure – Acceptance Criteria

How to applying acceptance criteria in project closure?

Steps:

- Verification: Ensure that all deliverables meet the acceptance criteria.
- Client Review: Present deliverables to the client for review against the criteria.
- Formal Sign-Off: Obtain formal approval from the client once the acceptance criteria are met.

Outcome:

- Successful Closure: The project is considered successfully closed when the client accepts the deliverables.
- Next Steps: Archive project documents, release resources, and conduct a post-project review.

More examples will be practiced in this weeks worshop ☺

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Any Questions?



Image Source:[Top 50 UX Design Memes on the Internet," Uxeria]